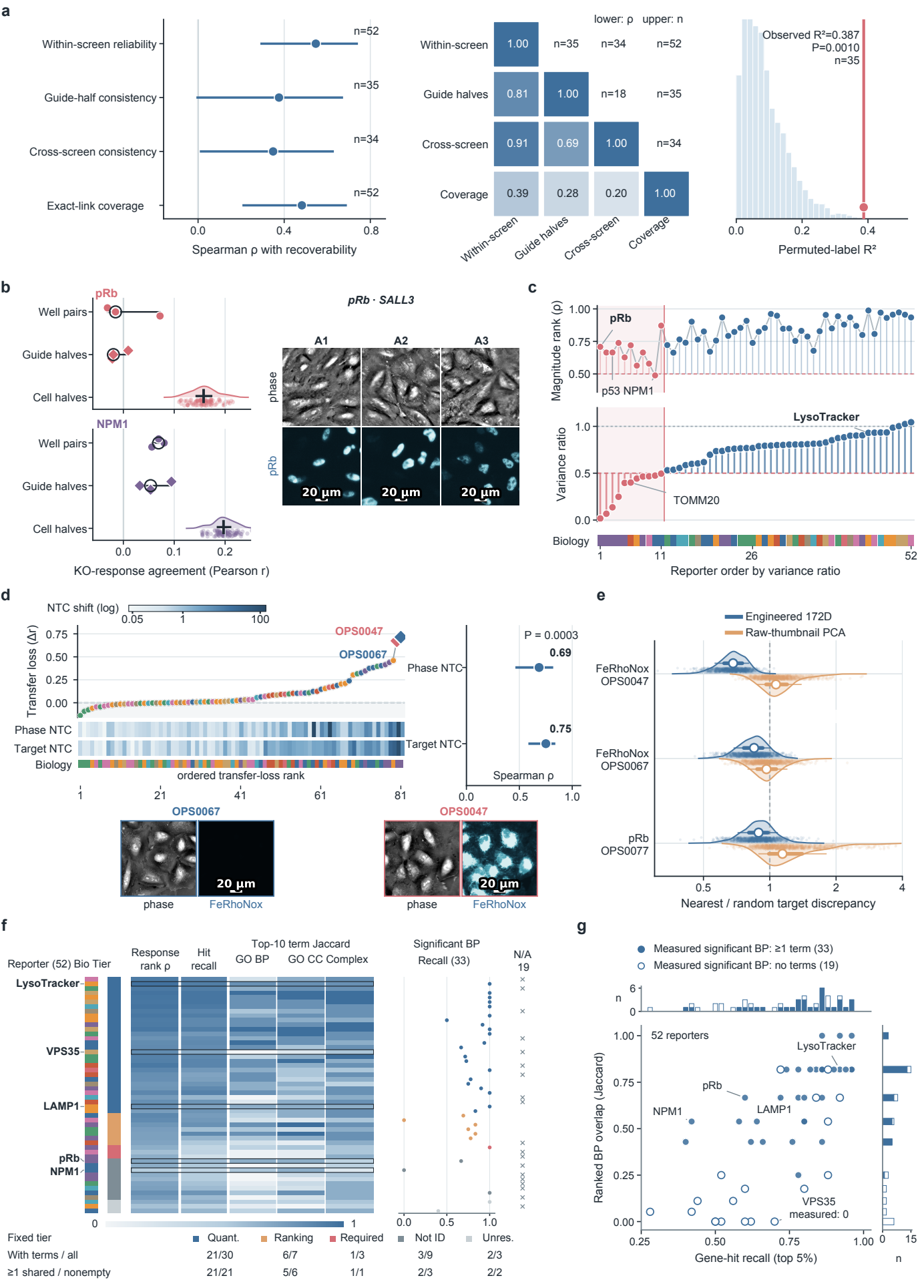

a
Within-screen reliability
Guide-half consistency
Cross-screen consistency
Exact-link coverage
n=52
n=35
n=34
n=52
0.0
0.4
0.8
Spearman ρ with recoverability
lower: ρ upper: n
Within-screen
Guide halves
Cross-screen
Coverage
1.00
n=35
n=34
n=52
0.81
1.00
n=18
n=35
0.91
0.69
1.00
n=34
0.39
0.28
0.20
1.00
Observed R²=0.387
P=0.0010
n=35
0.0
0.2
0.4
Permuted-label R²
b
pRb
Well pairs
Guide halves
Cell halves
NPM1
Well pairs
Guide halves
Cell halves
0.0
0.1
0.2
KO-response agreement (Pearson r)
pRb · SALL3
A1
A2
A3
phase
pRb
20 µm
20 µm
20 µm
c
Magnitude rank (ρ)
1.00
0.75
0.50
pRb
p53 NPM1
Variance ratio
1.0
0.5
0.0
LysoTracker
TOMM20
Biology
1
11
26
52
Reporter order by variance ratio
d
NTC shift (log)
0.05
1
100
Transfer loss (Δr)
0.75
0.50
0.25
0.00
OPS0047
OPS0067
Phase NTC
Target NTC
Biology
ordered transfer-loss rank
1
21
41
61
81
P = 0.0003
0.69
Phase NTC
0.75
Target NTC
Spearman ρ
0.0
0.5
1.0
OPS0067
20 µm
phase
FeRhoNox
OPS0047
20 µm
phase
FeRhoNox
e
Engineered 172D
Raw-thumbnail PCA
FeRhoNox
OPS0047
FeRhoNox
OPS0067
pRb
OPS0077
0.5
1
2
4
Nearest / random target discrepancy
f
Reporter (52) Bio Tier
Response rank ρ
Hit recall
Top-10 term Jaccard
GO BP
GO CC Complex
Significant BP
Recall (33)
N/A
19
LysoTracker
VPS35
LAMP1
pRb
NPM1
0
1
0.0
0.5
1.0
Fixed tier
Quant.
Ranking
Required
Not ID
Unres.
With terms / all
21/30
6/7
1/3
3/9
2/3
≥1 shared / nonempty
21/21
5/6
1/1
2/3
2/2
g
Measured significant BP: ≥1 term (33)
Measured significant BP: no terms (19)
n
6
0
52 reporters
LysoTracker
pRb
NPM1
LAMP1
VPS35
measured: 0
Ranked BP overlap (Jaccard)
1.00
0.75
0.50
0.25
0.00
0.25
0.50
0.75
1.00
Gene-hit recall (top 5%)
0
15
n